

Bahir Dar Institute of Technology

Faculty of Electrical and Computer Engineering

Assignment Questions

Chapter 1: Transformers (2 Questions)

Q1. A 1-phase 230/115 V transformer delivers 10 A to the secondary at unity power factor.

The equivalent resistance and reactance referred to the primary are $0.6\ \Omega$ and $1.2\ \Omega$ respectively.

If the core loss is 100 W, calculate:

1. The efficiency at full load,
2. The primary current and power factor at full load.

(Hint: Use approximate equivalent circuit and power flow concepts.)

Q2. Explain in detail how leakage reactance and core losses affect the voltage regulation and efficiency of practical transformers.

Then, derive an expression for percentage voltage regulation and explain the physical meaning of each term in your result.

(Your explanation must distinguish between leading and lagging power factors.)

Chapter 2: DC Machines (3 Questions)

Q1. A 250 V DC shunt generator delivers 50 A at 250 V. The armature and field resistances are $0.05\ \Omega$ and $125\ \Omega$ respectively.

Calculate: (i) The generated EMF, (ii) The efficiency of the generator if the stray losses are 1% of the output power,

and (iii) The speed at half-load, assuming flux remains constant.

Q2. Explain the voltage build-up process in a self-excited DC shunt generator.

Discuss the conditions necessary for successful voltage build-up and describe how each failure condition

(e.g., wrong polarity, loss of residual magnetism, high field resistance) affects generator performance.

Support your explanation with sketches of the OCC curve and field resistance line.

Q3. A 500 V, 4-pole DC shunt motor has a wave-connected armature with 600 conductors.

The armature and shunt field resistances are $0.2\ \Omega$ and $250\ \Omega$ respectively.

At full load, the line current is 50 A and the speed is 1200 rpm. Determine:

(i) The flux per pole, (ii) The torque developed, and (iii) The speed when the torque is reduced to 60% of full-load torque

while flux decreases by 4%. (Assume negligible armature reaction and brush drop.)

Chapter 3: Synchronous Machines (3 Questions)

Q1. Derive the expression for the electromagnetic power developed by a cylindrical-rotor synchronous generator

in terms of E_f , V_t , X_s , and the load angle δ . Then explain how power varies with δ and discuss the physical significance of maximum power angle.

Q2. A 3-phase, 11 kV, 100 MVA, 50 Hz, star-connected synchronous generator has a synchronous reactance of 1.2Ω per phase

and negligible resistance. Determine: (i) The internal generated EMF per phase, and (ii) The voltage regulation at 0.8 power factor lagging and leading.

(Sketch the phasor diagrams for both cases.)

Q3. Explain why a synchronous motor is not self-starting and describe three practical methods used to start it.

Then, discuss how power factor control is achieved in synchronous motors and how it affects system voltage stability when operating in parallel with other machines.

Prepared for: Electrical Machines Course

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